

Alma Ghafari

Khoury College of Computer Sciences, Northeastern University, Boston, MA

[LinkedIn](#) • [Google Scholar](#)

EDUCATION

- 2023 – present **Ph.D. in Computer Science**, Northeastern University, Boston, MA
Advisor: Soheil Behnezhad
- 2023 – 2025 **M.Sc. in Computer Science**, Northeastern University, Boston, MA
Advisor: Soheil Behnezhad
- 2019 – 2023 **B.Sc. in Computer Engineering**, Sharif University of Technology, Iran

HONORS & AWARDS

- 2023 **Khoury Distinguished Fellowship**, Northeastern University, one-year fellowship toward Ph.D.
- 2018 **Gold Medal**, 37th Iranian National Mathematical Olympiad

PUBLICATIONS

Author order is alphabetical. Publications prior to 2023 appear under M. Ghafari.

CONFERENCE & JOURNAL PAPERS

- [1] **Markov Chains with Rewinding**
Amir Azarmehr, Soheil Behnezhad, **A. Ghafari**, Madhu Sudan. [\[Random 2026\]](#)
- [2] **Lower Bounds for Non-adaptive Local Computation Algorithms** [\[FOCS 2025\]](#)
Amir Azarmehr, Soheil Behnezhad, **A. Ghafari**, Madhu Sudan.
- [3] **Stochastic Matching via In-n-Out Local Computation Algorithms** [\[STOC 2025\]](#)
Amir Azarmehr, Soheil Behnezhad, **A. Ghafari**, Ronitt Rubinfeld.
- [4] **Correlation Clustering Beyond the Pivot Algorithm** [\[ICML 2025\]](#)
Soheil Behnezhad, Moses Charikar, Vincent Cohen-Addad, **A. Ghafari**, Weiyun Ma.
- [5] **Fully Dynamic Matching and Ordered Ruzsa–Szemerédi Graphs** [\[FOCS 2024\]](#)
Soheil Behnezhad, **A. Ghafari**.
- [6] **Flips in Two-Dimensional Hypertriangulations** [\[EJC 2022\]](#)
Herbert Edelsbrunner, Alexey Garber, **M. Ghafari**, Teresa Heiss, Morteza Saghafian.
- [7] **On Angles in Higher-Order Brillouin Tessellations and Related Tilings in the Plane** [\[DCG 2022\]](#)
Herbert Edelsbrunner, Alexey Garber, **M. Ghafari**, Teresa Heiss, Morteza Saghafian.
- [8] **Brillouin Zones of Integer Lattices and Their Perturbations** [\[SIDMA 2022\]](#)
Herbert Edelsbrunner, Alexey Garber, **M. Ghafari**, Teresa Heiss, Morteza Saghafian, Mathijs Winttraecken.

MANUSCRIPTS UNDER SUBMISSION

- [8] **Caterpillar of Thoughts: The Optimal Test-Time Algorithm for Large Language Models**
Amir Azarmehr, Soheil Behnezhad, **A. Ghafari**.
- [9] **Streaming Algorithms for Monotonicity Testing**
Amir Azarmehr, Soheil Behnezhad, Lily Chung, **A. Ghafari**, Jane Lange, Ronitt Rubinfeld.

INVITED TALKS

- 2026 **Markov Chains with Rewinding and Application to test time compute in LLMs**
Algorithms Seminar, Google, Mountain View
- 2026 **Markov Chains with Rewinding**
RANDOM
- 2026 **Lower Bounds for Non-adaptive Local Computation Algorithms**
MIT Algorithms & Complexity Seminar
- 2026 **Lower Bounds for Non-adaptive Local Computation Algorithms**
Brown University Theory Seminar
- 2025 **Lower Bounds for Non-adaptive Local Computation Algorithms**
Sublinear Algorithms Workshop
- 2024 **Dynamic Matching and Ordered Ruzsa–Szemerédi Graphs**
Northeastern Theory Seminar
- 2024 **Upper Bounds on Ordered Ruzsa–Szemerédi Graphs**
Sublinear Graph Simplification Workshop, Simons Institute for the Theory of Computing
- 2024 **Sublinear Algorithms for Correlation Clustering: Beyond Pivot**
Old Questions and New Directions in the Theory of Clustering Workshop, UCSD
- 2021 **Brillouin Zones of Integer Lattices and Their Perturbations**
Institute of Science and Technology Austria (ISTA)

RESEARCH VISITS & INTERNSHIPS

- 2024 **Simons Institute for the Theory of Computing**, UC Berkeley
Host: Soheil Behnezhad
- 2022 **Max Planck Institute for Informatics**, Saarbrücken, Germany
Hosts: Joachim Spoerhase, Danupon Nanongkai
- 2021 – 2022 **Institute of Science and Technology Austria (ISTA)**, Klosterneuburg, Austria
Hosts: Morteza Saghaian, Herbert Edelsbrunner

TEACHING

- 2026 Teaching Assistant, *Theory of Computation* (undergraduate), Northeastern University
- 2024 Teaching Assistant, *Algorithms* (undergraduate), Northeastern University
- 2018 – 2020 Instructor, Mathematical Olympiad preparation in Combinatorics and Number Theory, Farzanegan High School
- 2019 Problem-setter and instructor, [Iranian National Mathematical Olympiad](#)
- 2019 Instructor, [Iranian Geometry Olympiad](#)

INDUSTRY EXPERIENCE

- 2020 – 2021 **Quantitative Analyst**, [Tosan Ofogh Hazare](#)
Designed trading algorithms and analyzed market data using financial and probabilistic models for the London Stock Exchange; co-developed a real-time trading platform.
Mentors: Mohammad Hossein Seyedsalehi, Mahdi Moosavi.

PROFESSIONAL SERVICE

- 2026 **Gold Reviewer**, ICML 2026
- 2024 – 2026 **Conference reviewing**: SODA '25, ICALP '25, FOCS '25, EC '25, SODA '26, SODA '26, ITCS '26, FOCS '26, NeuRIPS '26, WINE '26, SODA '27

2025	Poster, <i>Fully Dynamic Matching and Ordered Ruzsa–Szemerédi Graphs</i> WALDO
2024	Posters, <i>Fully Dynamic Matching and Ordered Ruzsa–Szemerédi Graphs</i> WOLA, TOCA-SV
2022	Attendee, Summer School in Combinatorial Optimization, EPFL

TECHNICAL SKILLS

Programming: Python, Java, C, C++, R, Qiskit, pytorch • **Graduate coursework:** Quantum Computing, Machine Learning, Statistical Methods for Computer Science, Advanced Algorithms